

Exam policy:

- *The exam is 50 minutes (9:00–9:50 AM) and is closed book, closed notes.*
- *You are allowed one hand-written cheat sheet of at most one two-sided page. Please hand in your cheat sheet with your solutions.*
- *Justify all your answers. It is not enough to just have the right answer. You must also convince us that you got the right answer for the right reasons.*
- *No calculators.*

Each of the three questions is worth an equal number of points.

The exam questions are on the other side of this page.

Please do not turn over this page until you are told to start the exam.

Good luck!

Disclaimer: some problems on this midterm may not be historically accurate.

Q. 1 (Plagues). The passover feast, presently ongoing, commemorates the liberation of the Israelites from slavery in Egypt (some time ago). The Egyptian Pharaoh was somewhat reluctant to comply with Moses' demand to let them go. To help persuade the Pharaoh, God sent a number of pretty nasty plagues his way.

- a. Plagues arrive as a Poisson process with rate 1 plague/week. After each plague, the Pharaoh first agrees to let the Israelites go but then reneges on his promise with probability 0.9; only with probability 0.1 are they actually allowed to leave. The Pharaoh's actions after each plague are independent. What is the distribution of the time it takes before the Israelites finally leave Egypt?
- b. There are 10 different types of plagues (blood, frogs, lice, etc.) Every time a plague is sent, the plague type is selected independently (of each other and everything else) with equal probability from the 10 types. What is the probability that each of the 10 types of plagues is sent exactly once before the Israelites leave Egypt?

Q. 2 (Matzot). Once the Israelites finally got the word that they could leave Egypt, they had to hurry up and get out within 12 hours. That did not leave much time to bake bread. The bread making process involves two long stages: first a rising stage, and then a proofing stage. The rising and proofing stages take independent, exponentially distributed amounts of time with expected lengths 8 and 4 hours, respectively. Unless both stages are completed, the bread will end up flat and unappetizing (a "Matzoh").

- a. What is the expected time taken by the longer of the two stages? (That is, compute the expected maximum of the rising time and the proofing time.)
- b. What is the probability that the whole process (i.e., first rising and then proofing) could be completed within the given 12 hours?

Q. 3 (In the desert). Once the Israelites leave Egypt, they are destined to wander around randomly in the wilderness. We model the wilderness by dividing it into segments of increasing distance to Egypt: segment 0 is Egypt; 1 is right across the border from Egypt; 2 is Mount Sinai; 3 is Kadesh; and 4 is the land of Israel.

The Israelites wander the wilderness according to a random walk. They start at location 1 right across the border from Egypt. Every year, they move one step closer to Israel with probability $\frac{2}{30}$; one step closer to Egypt with probability $\frac{1}{30}$; and remain in the same location with probability $\frac{27}{30}$. The Israelites continue wandering until they either reach Israel or unfortunately find themselves back in Egypt.

- a. What is the probability that the Israelites reach Israel in exactly 4 years?
- b. What is the expected number of years the Israelites wander in the wilderness?